

**Designing Artificial Intelligence to Support Rigorous
Mathematics Learning
for Students from Underprivileged Communities**

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Abstract

The rapid expansion of generative artificial intelligence (AI) in educational settings has prompted significant debate about whether these tools have the potential to reduce or exacerbate existing inequities in student learning. This study examines two related questions: how AI-driven learning systems can be designed to assess prior knowledge and scaffold rigorous mathematics instruction for students from low-income communities, and to what extent current AI educational tools reproduce existing opportunity gaps rather than address them. Grounded in sociocultural theory and critical pedagogy, the study adopts a qualitative case study design to investigate these questions in a Title I school where generative AI tools are already in use in mathematics classrooms. Data would be collected through semi-structured interviews with students and teachers, classroom observations conducted over six to eight weeks, and document analysis of AI-generated mathematical tasks and student work. Data would be analyzed using thematic analysis following Braun and Clarke's (2006) six-phase approach. Anticipated results suggest that current AI tools are likely to generate low-cognitive-demand mathematical content, reproduce deficit-oriented narratives about marginalized students, and benefit more mathematically prepared students disproportionately — patterns that would confirm and extend existing scholarship on generative AI's limitations in equity-centered education. However, teacher mediation is expected to emerge as a critical moderating factor, with educators who use AI strategically and relationally producing more equitable outcomes. The implications of these findings point toward the urgent need for equity-centered design guidelines for AI tools in mathematics education, as well as professional development frameworks that equip teachers to use AI in ways that genuinely serve historically underserved learners.

Keywords: generative AI, mathematics education, educational equity, underprivileged students, qualitative case study, sociocultural theory

Introduction

Students from low-income and historically underserved communities continue to face persistent disparities in access to rigorous mathematics education — disparities rooted not in individual deficiencies but in systemic inequities including underfunded schools, high teacher turnover, and curricula that routinely underestimate students' mathematical potential (Walkington, 2025; Price et al., 2024). The recent emergence of generative artificial intelligence tools, particularly large language models such as ChatGPT, has generated significant optimism about their potential to personalize instruction and democratize access to high-quality educational support (Ahmad et al., 2023). Yet growing evidence suggests that the uncritical adoption of these tools may reproduce — and in some cases deepen — the very inequities they are presumed to address (Sapkota & Bondurant, 2024; Price et al., 2024). Despite the rapid pace of AI integration in education, research focused specifically on how generative AI can be designed to serve underprivileged mathematics learners remains strikingly limited. This study addresses that gap by examining, through a qualitative case study, how AI tools currently function in a school serving low-income students and what design principles might support more equitable outcomes.

Literature Review

Theoretical Framework

This study is grounded in two complementary theoretical traditions. Drawing on Vygotsky's sociocultural theory, mathematical learning is understood as a fundamentally social and context-dependent process, in which effective scaffolding must begin from students' existing knowledge and community experiences rather than treating them as blank slates. Critical pedagogy, extending Freire's foundational work into mathematics education, frames this study's concern with how educational technologies can perpetuate or disrupt systemic inequities. Together, these frameworks

position AI not as a neutral tool but as a socially embedded artifact whose design choices carry real consequences for marginalized learners.

Generative AI in Mathematics Education

Scholarship on generative AI in mathematics education reflects a field at an inflection point. Walkington (2025) provides a comprehensive review identifying four key domains of AI application: problem-solving support, tutoring and feedback, task adaptation, and teacher planning. Ahmad et al. (2023) characterize large language models as a landmark development in conversational AI and examine the wide range of institutional responses — from outright bans to enthusiastic adoption — as schools navigate this new landscape. Matzakos et al. (2023) further argue that LLMs offer functional and pedagogical advantages over traditional computer algebra systems, potentially prompting a rethinking of mathematics curriculum and assessment in higher education.

However, critical scrutiny has begun to complicate this optimistic picture. Bui et al. (2024), in a scoping review of 28 studies, found that while ChatGPT performs adequately on procedural tasks, it struggles with complex, multi-step problems requiring visual or spatial reasoning — precisely the kinds of tasks central to rigorous mathematical thinking. Sapkota and Bondurant (2024) corroborated this finding, showing that ChatGPT-generated mathematics tasks defaulted to procedural, low-cognitive-demand formats regardless of how prompts were varied, and contained conceptual inaccuracies. Kaya and Yavuz (2025) offer a more nuanced view, examining the conditions under which AI tools might support higher-order mathematical reasoning, though they acknowledge significant variability in outcomes.

Equity and AI in Education

The equity dimensions of AI integration are among the most urgent and least resolved areas of the literature. Price et al. (2024) documented troubling patterns in ChatGPT's outputs, finding that the tool reproduced racial stereotypes in descriptions of mathematical learners — framing White and Asian students as the unmarked norm while characterizing Black students as exceptional. Thomas et al. (2025) offer a more hopeful counterpoint, demonstrating that advanced LLMs can assess tutors' equity competencies with meaningful accuracy, suggesting potential for AI-assisted equity-focused professional development. Smith et al. (2024) extend the conversation further by showing that AI tools can generate useful mathematical writing models for elementary students with learning disabilities, pointing toward possibilities for more targeted AI applications for specific underserved populations.

Knowledge Gaps

Despite the growing volume of scholarship on AI in mathematics education, critical gaps remain. No studies to date have systematically examined how generative AI can be designed to build on the prior knowledge and community assets of students from low-income communities. The literature also lacks research on hybrid human-AI instructional models that combine the scalability of AI with the relational dimensions of skilled human teaching — a gap especially relevant for students whose academic success depends on strong human connections. Ethical questions surrounding algorithmic bias, data privacy, and the political economy of AI adoption in under-resourced schools remain largely unaddressed. This study is designed to contribute to filling these gaps.

Method

Research Design and Paradigm

This study adopts a qualitative case study design grounded in an interpretivist paradigm. Rather than seeking to measure outcomes or test hypotheses statistically, the goal is to generate rich, contextually grounded understanding of how AI tools are experienced within a specific school community. Case study methodology is appropriate here because both research questions — asking how AI can be designed to scaffold rigorous mathematics learning, and to what extent current tools reproduce opportunity gaps — require sustained, in-depth examination of a bounded real-world phenomenon rather than broad statistical generalization.

Participants and Recruitment

Participants would be recruited through purposive sampling from a Title I elementary or middle school in a low-income urban community where generative AI tools are already in use in mathematics instruction. Contact would be made with the school principal to request access, followed by IRB approval prior to any data collection. Approximately 2 to 4 mathematics teachers who actively integrate AI tools in their instruction and 6 to 10 students in grades 4 through 8 would be selected. Written informed consent would be obtained from all adult participants, and parental consent along with student assent forms would be collected for all minor participants.

Data Collection

Data would be collected over approximately six to eight weeks using three methods. First, semi-structured interviews would be conducted with each teacher (60–90 minutes) and each student participant (30–45 minutes), exploring experiences with and perceptions of AI tools in mathematics learning. All interviews would be audio-recorded and transcribed verbatim. Second, classroom observations (2–3 per week, 45–60 minutes each) would be conducted during mathematics lessons where AI tools are in use, with detailed field notes documenting student-AI interactions, task quality, and equity-relevant patterns. Third, document analysis would be conducted on

AI-generated mathematical tasks, student work samples, and school policies governing AI use. This triangulation of data sources is appropriate because it allows the researcher to cross-verify findings and develop a more complete picture of the phenomenon than any single source could provide.

Data Analysis

Data would be analyzed using thematic analysis following Braun and Clarke's (2006) six-phase approach: familiarization with the data, initial coding, theme development, theme review, theme definition and naming, and writing up. Trustworthiness would be established through triangulation across data sources, member checking with teacher participants, maintenance of an audit trail, reflexive journaling, and thick description of the research context.

Anticipated Results

Based on the literature reviewed, this study anticipates a mixed picture in which AI tools offer some genuine affordances for students while also reproducing a number of the equity concerns identified in prior research. With respect to the first research question, current AI tools are expected to fall short of adequately assessing or building on the prior knowledge and cultural backgrounds of underprivileged students. AI-generated mathematical tasks are anticipated to trend toward procedural, low-cognitive-demand content — consistent with Sapkota and Bondurant (2024) — and teachers are expected to report supplementing AI-generated materials with their own, suggesting that human mediation is a necessary condition for meaningful AI-assisted learning.

Regarding the second research question, the study anticipates finding evidence that AI tools do, to some degree, reproduce existing opportunity gaps. Document analysis may reveal deficit-oriented patterns in AI-generated feedback consistent with Price et al.

(2024). Student interviews are expected to show that those with stronger prior mathematical foundations benefit more from AI scaffolding than those with greater foundational gaps — a pattern mirroring broader inequities in educational technology effectiveness. Nonetheless, teachers who deploy AI strategically and relationally are expected to produce more equitable outcomes, pointing to teacher practice as a potentially important lever for equity.

Anticipated Discussion

The anticipated findings largely confirm what the literature has begun to suggest about generative AI's limitations in equity-centered mathematics education. The expected pattern of low-cognitive-demand AI content aligns with Sapkota and Bondurant (2024) and Bui et al. (2024), and the anticipated differential benefit across student groups confirms Bui et al.'s (2024) warning that AI may widen achievement gaps if left unmediated. The anticipated presence of deficit-oriented framings in AI outputs aligns with Price et al. (2024), suggesting these biases may manifest not only in explicit descriptions but also in how AI responds to and scaffolds student work.

At the same time, these findings would disconfirm the broader techno-optimist framing advanced by Walkington (2025) and Ahmad et al. (2023) — at least in the specific context of underprivileged learners — by demonstrating that AI's benefits are unevenly distributed and contingent on human mediation. This extends the literature in an important direction: rather than asking whether AI works, the more productive question becomes for whom and under what conditions it works. The anticipated finding regarding teacher mediation as a moderating factor represents a gap the current literature has not adequately addressed, and points toward the importance of professional development frameworks that support equity-centered AI integration.

Probable Conclusion

This study examined how AI-driven learning systems can be designed to scaffold rigorous mathematics instruction for students from low-income communities, and to what extent current AI tools reproduce rather than address existing opportunity gaps. Drawing on a qualitative case study approach grounded in sociocultural theory and critical pedagogy, anticipated findings suggest that current generative AI tools are likely to generate low-demand content, benefit more mathematically prepared students disproportionately, and embed subtle deficit-oriented narratives in their outputs — collectively representing a reproduction of existing inequities. Teacher mediation emerged as a potentially important moderating factor, suggesting that the relationship between AI and equity is not fixed but is shaped by the instructional decisions of the educators who deploy these tools.

The significance of this research lies in its focus on a population and a set of questions that the field has largely overlooked as AI adoption accelerates. Several limitations should be acknowledged: the single-site case study design limits statistical generalizability; the findings remain predictive rather than empirically confirmed; and the researcher's positionality may shape data interpretation despite reflexivity strategies. Future research should expand this inquiry across multiple school sites and geographic contexts, investigate the specific AI design features that promote or hinder equitable outcomes, and explore what professional knowledge enables teachers to use AI in genuinely equity-centered ways. Ultimately, this study contributes to a growing body of work insisting that the central question in AI-assisted education is not whether AI works, but how it can be designed and deployed to work for everyone.

Timeline

The following timeline outlines the proposed sequence of activities for carrying out this qualitative case study. The total projected duration of the study from IRB approval

through final report submission is approximately seven months.

Month 1 — IRB Approval and Site Access: Submit the study protocol to the Institutional Review Board for ethical review and approval. Simultaneously initiate contact with a Title I school principal or district administrator to request site access and establish a working relationship with the school community.

Month 2 — Participant Recruitment and Consent: Once IRB approval is obtained and site access is confirmed, distribute informed consent forms to teacher participants and parental consent and student assent forms to identified student participants. Finalize the participant pool and schedule initial interviews.

Months 3–4 — Data Collection: Conduct semi-structured interviews with all teacher and student participants. Carry out classroom observations (2–3 per week) over the six-to-eight-week observation window. Collect relevant documents including AI-generated tasks, student work samples, and school policy documents throughout this period.

Month 5 — Data Analysis: Transcribe all interview recordings. Apply thematic analysis across all three data sources following Braun and Clarke's (2006) six-phase process. Conduct member checking with teacher participants to verify emerging interpretations and refine the thematic structure accordingly.

Month 6 — Writing and Revision: Draft the full research report including all sections from introduction through conclusion. Incorporate feedback from member checking and reflexive journaling. Review the manuscript for internal consistency, APA formatting, and alignment with the research questions.

Month 7 — Final Submission: Complete final revisions, proofread the manuscript, and submit the finished research report for review and dissemination.

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